

The environmental impact of AI has been compared to everything, from scale to scale. But wouldn't it be useful to compare it to something like video games?

Posted by u/Questioner8297 • • 0 points • 28 comments

"From our metering, we find that although some consoles are left on nearly 24 h/day, the overall average usage is lower than many other studies have assumed, leading to a US national energy consumption estimate of 7.1 TWh in 2012."

<https://eta-publications.lbl.gov/publications/video-game-console-usage-and-us>

Video game in general (2019):

"In aggregate, we find that gaming represents \$5 billion per year in energy expenditures across the United States or 34 TWh/year (2.4% of residential electricity nationally), with 24 MT/year of associated carbon-dioxide emissions equivalent to that of 85 million refrigerators or over 5 million cars"

[<https://eta.lbl.gov/publications/toward-greener-gaming-estimating>]
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About data center.

"In its simplest form, a data center is a physical facility that houses and runs large computer systems. U.S. data center annual energy use in 2023 (not accounting for cryptocurrency) was approximately 176 terawatt-hours (TWh), approximately 4.4% of U.S. annual electricity consumption that year, according to a report by Lawrence Berkeley National Laboratory. A data center typically contains multiple computer servers, data storage devices, and network equipment that can provide information technology (IT) infrastructure service for organizations to store, manage, process, and transmit large amounts of data. Some projections show that data center energy consumption could double or triple by 2028, accounting for up to 12% of U.S. electricity use."

<https://www.congress.gov/crs-product/R48646>

As you can see, we are wasting a huge amount of energy, but for some reason people are only worried about AI.

Comments

u/ChimpieTheOne • 1 points • 2025-12-26 23:44 UTC

Well, gaming is fun and benefits almost all.

AI (especially genAI) benefits mostly corporations and people who dgaf about the issues it comes with

u/HelpRespawnedAsDee • 1 points • 2025-12-26 20:37 UTC

When you bring this up, they deflect to "well at least games give something in return, what is AI giving me in return".

Which yes, is a laughable position.

u/xoexohehexox • 2 points • 2025-12-26 20:17 UTC

Last I read both global video game use (PC plus console plus mobile) and AI+crypto together were both somewhere around 350-400 TWh which is small potatoes, global consumption is something like 23000 TWh, the big ticket items are HVAC and fuel refining I believe.

u/Quirky-Complaint-839 • 6 points • 2025-12-26 13:33 UTC

The Internet is run by data centers. Either businesses run their computing on premise or in data centers. If on premise, it means physical offices. It means that people are commuting to work. Not the commute is factored in.

One thing isn't just one thing.

u/Constant_Topic_1040 • -1 points • 2025-12-26 13:28 UTC

AI data centers are getting so big that they need their own energy infrastructure, as they can no longer draw on the existing infrastructure. Economically that should help out, but let's not act like there isn't a problem, that will require lots of energy usage

u/Familiar-Art-6233 • 5 points • 2025-12-26 20:01 UTC

AI isn't intrinsically tied to datacenters. My laptop generating images isn't consuming water or requiring dedicated energy infrastructure, that's just silly

u/Emergency-Name-2334 • 4 points • 2025-12-26 18:56 UTC

That's not actually true, BTW. These larger data centers *are* absolutely being built, yes, but it's not based on current AI usage, it's based on companies that are convinced AGI will happen (which, even as someone who is extremely pro-AI, I remain *extremely* skeptical will be achieved in our lifetimes, assuming it's even possible at all), and want to be the first to market with the infrastructure to support it.

Its economic speculation. They think a similar, related technology is about to exist, and they are trying to race to be the first one there. But that doesn't actually mean that technology is there, and these data centers will likely be scrapped and repurposed with their parts collapsing the secondary market if AGI does not happen quickly (which, again, we still don't even know if it will be definitively possible)

Anyone who wants to be an accelerationist towards the AI bubble bursting SHOULD be encouraging these companies to invest so much they collapse under their own overhead costs, actually.

u/EvilKatta • 9 points • 2025-12-26 14:51 UTC

Data centers isn't an AI only thing, and even the AI they do isn't just image and text generators. What is an AI specific problem here, exactly?

u/Constant_Topic_1040 • -2 points • 2025-12-26 14:58 UTC

I'm not even gonna take you seriously cuz I didn't say anything about things it generates. That's weird baiting for an argument. Also I didn't say that it was an AI specific problem, I said that there are a lot of logistic issues that you can't hand wave away, and people can be worried about New energy usage, in which infrastructure has to be built to accommodate

u/EvilKatta • 9 points • 2025-12-26 15:04 UTC

Exactly, I don't understand the issues you're trying to raise. It's not an AI specific issue, so we would run into it regardless.

If there weren't image and text generators, there would still exist search engines, big data analyzers etc. They existed before 2022, and there wasn't an outrage about their energy usage. (A lot of current outrage is explained by "But AI is useless", but people didn't care before 2022.)

Is requiring new energy infrastructure an issue? We as society are in the process of migration to sustainable energy sources, so we would also require it regardless.

u/Constant_Topic_1040 • -1 points • 2025-12-26 15:12 UTC

People were completely ignorant of it and didn't understand the scale. If they understood it when the first data centers were made, they probably would've reacted a similar way. Especially when data centers as a whole are poised to double or triple. Especially before everything used the internet people were absolutely clueless about how it worked and still are. Now a days people are on average more understand of technology compared to 20 years ago.

So as I said, the people who live near these new data centers have legitimate concerns, and hand waving it away is not helpful. Especially this post which compares gaming as a whole, spread out over

thousands of energy grids, and is just over half of the current data center usage which will double or triple

u/_Brick_- • -5 points • 2025-12-26 13:08 UTC

AI is bad games are good

u/Capital_Pension5814 • 9 points • 2025-12-26 13:34 UTC

Soooo...the environmental argument is null

u/_Brick_- • -6 points • 2025-12-26 13:34 UTC

Nah....im just too lazy to care

u/Human_certified • 18 points • 2025-12-26 12:47 UTC

Yup. For the first time ever, people are realizing that "the cloud" is actually a power-intensive industry, and some abstraction (and has been for 20-25 years). To make the panic even less relevant:

\- Electricity consumption $\neq$ energy consumption (electricity accounts for about a quarter).

\- The 12% figure is from extremely disreputable sources who don't understand GPUs and are bad at math.

\- Any figures that talk about "US power consumption" are also suspect, because AI is a global industry, but its data centers are disproportionately in the US.

\- Data centers are always vastly more efficient than distributed, local computing. They're just more visible.

\- People love to talk about "but the training costs!!!" as if they're a constant hidden factor. What they are is CapEx, one-off. So the more a model is used, the lower the training costs per inference run become. Also, not every model upgrade is a massive pretraining run. That's how OpenAI is able to push out GPT-5, 5.1, 5.2 in rapid succession. (FWIW, people have been screaming that training GPT-3.5 took "as much energy as 160 US households use in a year!" - yeah, that's about the same as manufacturing a handful of small cars).

\- Gaming, streaming, music, etc. all provide entertainment. AI provides entertainment *at the very minimum*. That is, the uses of AI that many people think are the biggest "waste", can be categorized as entertainment. However, AI also provides massive business value, research value, time saved and practical advice to consumers, etc. So we'd have to pull the plug on gaming, streaming, social media, etc. long before we did the same to AI.

u/fleegle2000 • 7 points • 2025-12-26 11:46 UTC

My most charitable or "steelman" take on the anti-AI position is that A) AI is a waste of computer resources, as compared to even gaming, which at least brings a lot of joy to people, and can be a source of income for some (so can AI but they would argue on the backs of artists, so not legitimately), and B) while submitting prompts is relatively benign, we have to consider that it can take months of continuous computation to train the models, which needs to be factored in to the energy cost.

I remain somewhat on the fence about A; though I disagree that making money off AI art is unethical, I do question ultimately how useful genAI will be to society and wonder if the cons outweigh the pros, like social media. I remain unconvinced on B and would need to see some hard data to make up my mind. I think that right now a lot of resources are being poured into AI but I also know that some models are

heavily used so the months of training are worth the energy cost even if it is higher than I imagine. I could see a case being made against spending a lot of energy on a model that barely gets used (either because it is unpopular or is superceded by another model shortly thereafter).

u/xoexohexox • 1 points • 2025-12-26 20:19 UTC

More than half of LLM use is companionship, call it entertainment but it's more than that. The bulk of the rest is computer programming and that has really democratized how people use computers. I get projects up and running in days that would have taken me months before, it's neat. It's like 3d printing. Yeah maybe not everyone is making money off of doing it at an industrial scale, but I can like for example print out a perfect replacement for a missing knob on a device etc. Just costs feedstock and not shipping fees and fuel to deliver.

u/JunKazama2024 • 1 points • 2025-12-26 14:18 UTC

As I understand it the CO2 cost of training a ChatGPT version is similar to that of producing a single Hollywood film

u/Questioner8297 • 5 points • 2025-12-26 12:12 UTC

Honestly, I thought video and image generation models seemed closest to games. LLM, that is, text generation, is already efficient. At a minimum, searching with gpt 5.2 is quite good; it's not perfect, but it's quite convenient for finding information if you're not looking for a specific object, but rather a quick overview of the topic.

The biggest investments are in LLM. Models like the Nano Banana Pro combine both, making it difficult to even separate them.

Then, models like the nano banana pro can be used, with a bunch of caveats, to save diagrams/schemes that are useful for learning and for illustrating data.

Video generation models have remained somewhat of a separate entity. They consume a lot of power and are of little use. But isn't it possible to eventually create text, video, and images in a single model? It's entirely possible.

Additionally, as someone who enjoys LMM for entertainment, I can say that for me, it's a new type of entertainment, very different from everything else. It's like a computer game differs from a movie; the difference in interactivity is so vast that it creates a new meaning.